$A \in \mathbb{R}^{n \times n}$.

If the algebraic multiplicity of at least one eigenvalue λ_i is greater than its geometric multiplicity. λ_i is called defective eigenvalue and A is a defective matrix.

Def: A vector x_m is a generalized eigenvector of rank m of the matrix and corresponding to the eigenvalue λ if.

$$\begin{cases} (A-\lambda I)^m x_m = 0 \\ (A-\lambda I)^{m-1} x_m \neq 0 \end{cases}$$

Example:

$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\lambda = 1$ with $M = 2$.

$A - I\lambda = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \rightarrow v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $\gamma = 1$.

Note that, $(A - I\lambda) v_1 = 0$.

Let's have $(A - I\lambda) v_2 = v_1$,

$$\left(\begin{array}{cc|c} 0 & 1 & 1 \\ 0 & 0 & 0 \end{array}\right) \rightarrow v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

v_2 satisfies $\begin{cases} (A-\lambda I)^2 v_2 = (A - \lambda I) v_1 = 0 \\ (A - \lambda I) v_2 \neq 0 \end{cases}$

When solving ODE, we have,

$x'(t) = Ax$

If $m = 3$.

$x_3(t) = e^{\lambda t}(a(t) v_1 + b(t) v_2 + c(t) v_3)$ to let x_3 as a solution, we have $x_3'(t) = Ax_3$

Suppose $c(t) = 1$

$x_3'(t) = e^{\lambda t}(a(t)\lambda v_1 + b(t)\lambda v_2 + c(t)\lambda v_3) + e^{\lambda t}(a'(t) v_1 + b'(t) v_2)$ (1)

$$Ax_3 = e^{\lambda t}\left(a(t)(\lambda v_1) + b(t)(\lambda v_2 + v_1) + c(t)(\lambda v_3 + v_2)\right)$$

$$= \lambda e^{\lambda t}(av_1 + bv_2 + cv_3) + e^{\lambda t}(b(t)v_1 + c(t)v_2) \quad ②$$

① = ② s.t $\quad a'(t)v_1 + b'(t)v_2 = b(t)v_1 + c(t)v_2$

s.t suppose $c = 1$, $\quad b(t) = t$, $\quad a = \frac{t^2}{2}$

In general,

$$x_n = e^{\lambda t}\left(\frac{t^n}{n!}v_1 + \cdots + v_n\right)$$